Welcome

CS 472/572
Unsupervised Machine Learning

PCA Analysis

Week-9 -Class -2

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Topics

1. Eigenvectors and eigenvalues
2. Variance and multivariate Gaussian distributions
3. Computing a covariance matrix from data
4. Principal Components Analysis (PCA)

From Last Class Eigenvectors and eigenvalues

- Diagonal matrices have the property that they map any vector parallel to a standard basis vector into another vector along that standard basis vector.

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

eigenvalue equation

$$\Lambda \hat{e}_i = \lambda_i \hat{e}_i \quad , \quad i = 1,2 \ldots .n$$

- Any vector $\vec{v}$ that is mapped by matrix A onto a parallel vector $\lambda \vec{v}$ is called an eigenvector of A. The scale factor λ is called the eigenvalue of vector $\vec{v}$.

- A matrix in $\mathbb{R}^n$ has n eigenvectors and n eigenvalues.

Variance

- Variance is the sum of squares of differences between all numbers and means.

- Variance (σ^2) = (Sum of the squared differences from the mean) / (Total number of values)

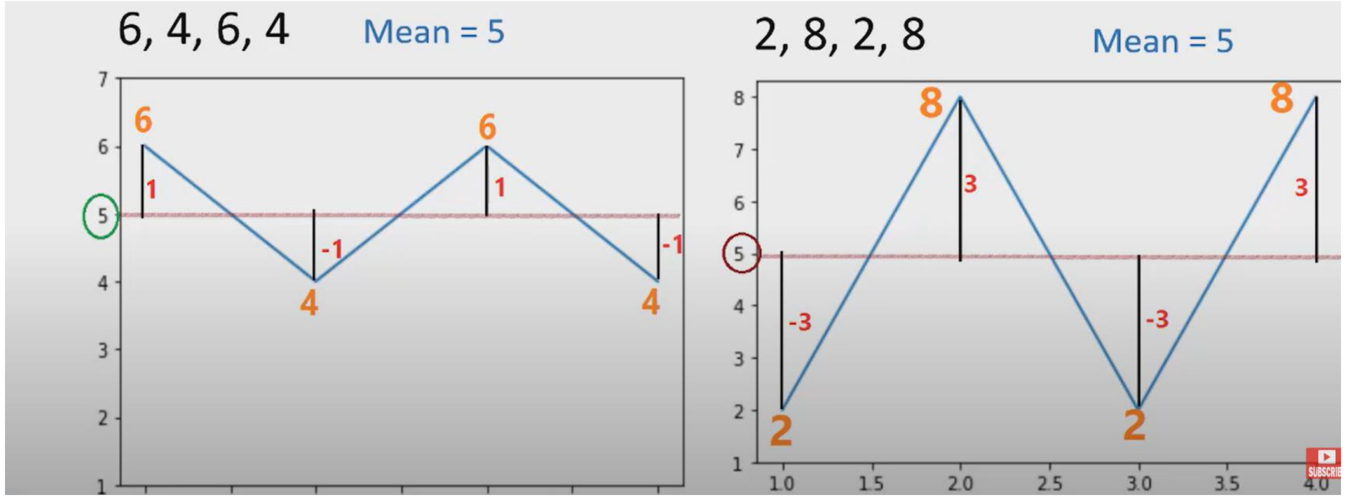
- In mathematical notation: $\sigma^2 = \Sigma(x - \mu)^2 / (n)$

Here:

- μ is the mean of independent features
- Mean (μ) = (Sum of all values) / (Total number of values)

Varience

- The variance is a measure that indicates how much data scatter around the mean



- In mathematical notation: $\sigma^2 = \Sigma(x - \mu)^2 / (n)$

6, 4, 6, 4 Mean = 5

Total Distance = $(6-5) + (4-5) + (6-5) + (4-5) = 0$ X

Total Distance = $(6-5)^2 + (4-5)^2 + (6-5)^2 + (4-5)^2 = 4$

Variance = $\dfrac{(6-5)^2 + (4-5)^2 + (6-5)^2 + (4-5)^2}{4} = 1$

Covariance

1. It is the relationship between a pair of random variables where change in one variable causes change in another variable.
2. It can take any value between -infinity to +infinity, where the negative value represents the negative relationship whereas a positive value represents the positive relationship.
3. It is used for the linear relationship between variables.
4. It gives the direction of relationship between variables.

Covariance

The formula for the covariance (Cov) between two random variables X and Y, each with N data points, is as follows:

$$Cov(X,Y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{X})(y_i - \bar{Y})$$

Where:

- Cov(X,Y) is the covariance between X and Y.
- N is the number of data points.
- Xi and Yi represent individual data points for X and Y, respectively.

Step 1: Obtain the Data

Collect and organise the raw dataset into a matrix of m observations × n variables

Data Matrix X

Arrange the data as a matrix X:

- Each row = one observation (sample)
- Each column = one variable (feature)

Dimensions: X is m × n

where m = number of observations
and n = number of variables

Matrix Layout

	Var 1	Var 2	...	Var n
Obs 1		x_{ij}	x_{ij}	x_{ij}
Obs 2		x_{ij}	x_{ij}	x_{ij}
...		x_{ij}	x_{ij}	x_{ij}

X → m rows (observations) × n cols (variables)

All variables must be numerical. Ensure data is complete before proceeding.

Step 2: Subtract the Mean

Centre the data by subtracting the mean of each variable from all its values

Why Centre the Data?

PCA finds directions of maximum variance.
If data is not centred, the covariance matrix measures variance around zero, not around the data's true centre.
Centring ensures we measure relative spread correctly.

Procedure

1. Compute mean μ_j for each variable j
2. Subtract: $\tilde{x}_{ij} = x_{ij} - \mu_j$ for all i, j

Formula

$$\mu_j = (1/m) \cdot \Sigma_i \, x_{ij}$$

$$\tilde{X} = X - \mu \quad \text{(centred data matrix)}$$

Effect:

Before
Mean ≠ 0

After
Mean = 0

After centring, mean of every variable in $\tilde{X}$ equals exactly zero

Step 3 : Calculate Covariance Matrix

Compute how every pair of variables varies together using the centred data

Covariance Formula

$$C = (1/(m-1)) \cdot \tilde{X}^T \cdot \tilde{X}$$

where $\tilde{X}$ is the centred data matrix

Each entry C_{ij} tells us:

- C_{ii} = variance of variable i
- C_{ij} = covariance of variables i and j

Result: C is an n × n symmetric matrix

Covariance Matrix Structure

$Var(X_1)$	$Cov(X_1,X_2)$	$Cov(X_1,X_3)$
$Cov(X_2,X_1)$	$Var(X_2)$	$Cov(X_2,X_3)$
$Cov(X_3,X_1)$	$Cov(X_3,X_2)$	$Var(X_3)$

= diagonal (variance terms)

= off-diagonal (covariance terms)

C is always symmetric: $C_{ij} = C_{ji}$ • Diagonal entries are always non-negative

Step 4 :Eigenvalues & Eigenvectors of C

Solve the characteristic equation $\det(C - \lambda I) = 0$ to find eigenvalues, then eigenvectors

Eigenvalues λ

$$\det(C - \lambda I) = 0$$

- Solve the characteristic equation
- Yields n eigenvalues: $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$
- Each λ = amount of variance in that direction

Eigenvectors v

$$(C - \lambda_i I) \cdot v_i = 0 \;\rightarrow\; \text{solve for each } v_i$$

Key Properties

1. **Eigenvalue = Variance**

   λ_i measures the variance of the data along eigenvector v_i. Larger λ = more important direction.

2. **Eigenvectors are Orthogonal**

   All principal component directions are perpendicular to each other — $v_i^T v_j = 0$ for $i \neq j$.

3. **Unit Eigenvectors**

   Normalise each eigenvector so $\|v_i\| = 1$. These become the principal component axes.

Sort eigenvalues in descending order: $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$ (most to least important)

Step 5: Choose Principal Components

Select the top k eigenvectors that capture sufficient variance for your application

How to Choose k

A **Variance Threshold**

Keep enough PCs to explain 95% (or 90%/80%) of total variance

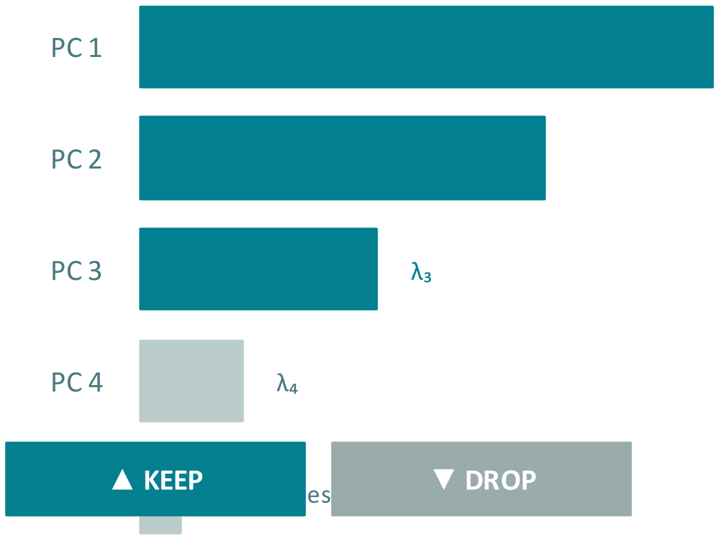
B **Scree Plot Elbow**

Plot eigenvalues vs PC number. Pick k at the 'elbow' where curve flattens

C **Fixed Dimension**

Set k to a target dimension (e.g. k=2 for 2D visualisation)

Eigenvalue Ranking



The top k eigenvectors form the projection matrix W of size n × k

Step 6 : Derive the New Dataset

Project the centred data onto the selected principal components to get the reduced representation

Projection Formula

$$Y = \tilde{X} \cdot W$$

$\tilde{X}$ = centred data matrix $(m \times n)$
W = projection matrix $(n \times k)$
Y = new reduced dataset $(m \times k)$

What It Means

Each row of Y is one observation expressed in the new PC coordinate system.

Dimension Reduction Flow

$\tilde{X}$	$\times$ W	= Y
m × n (original dims)	n × k (top-k eigenvectors)	m × k (reduced dataset)

Benefits:

- Variables in Y are uncorrelated (orthogonal)
- Storage: m·k instead of m·n values
- k << n means massive dimension reduction

Y has k columns — one per chosen principal component — and m rows (same as original data)

Step 7 : Analyse & Interpret Results

Examine the principal components and evaluate how well the reduced data represents the original

A Examine PC Loadings

Each eigenvector component (loading) tells you how strongly each original variable contributes to that PC. Large loading = strong influence.

B Variance Explained

Compute the proportion of variance each PC captures: $\lambda_i / \Sigma\lambda$. Plot the scree plot to visualise the distribution.

C Visualise Reduced Data

Plot Y on the first 2 or 3 PCs for visualisation. Clusters in PC space reveal underlying structure in the data.

D Reconstruction & Error

Reconstruct: $\hat{X} = Y \cdot W^T + \mu$. Compute reconstruction error $\|\tilde{X} - Y \cdot W^T\|^2$ to measure information lost.

The goal: determine whether k PCs faithfully represent the original n-dimensional data

PCA Algorithm: As a whole

1

Obtain Data

Arrange data as $m \times n$ matrix X

2

Subtract Mean

$\tilde{X} = X - \mu$ (centre each variable)

3

Covariance Matrix

$C = (1/(m-1)) \cdot \tilde{X}^T\tilde{X}$

4

Eigendecomposition

Solve $\det(C - \lambda I) = 0$, find v_i

5

Select k PCs

Choose top k by variance explained

6

Project Data

$Y = \tilde{X} \cdot W$ (new dataset $m \times k$)

7

Analyse Results

Loadings, variance %, reconstruction

Let's ………………

Solve a Numerical Example

Problem Definition

Objective: Reduce dimensions from 2D to 1D using PCA on a 4-sample dataset

Feature	Example 1	Example 2	Example 3	Example 4
X_1	4	8	13	7
X_2	11	4	5	14

2
Features
X_1 and X_2

4
Samples
Examples

1
Target Dims
After PCA

6
Steps
In this example

Step 1 : Calculate Mean

Compute the mean of each feature across all 4 samples

Formula

$\mu = (1/n) \cdot \Sigma x_i$

where n = number of samples
and x_i = individual values

Mean of X_1 = (4 + 8 + 13 + 7) / 4 = 32 / 4

$\mu_1 = 8$

Calculation

Mean of X_2 = (11 + 4 + 5 + 14) / 4 = 34 / 4

$\mu_2 = 8.5$

Result Summary

Mean vector: $\mu = [\,8,\ 8.5\,]$

Step 2 : Covariance Matrix

Measure how the two features vary together using centered data

Covariance Formula

$\mathrm{Var}(X_1) = \Sigma(x_{1i} - \mu_1)^2 / (n-1)$
$\mathrm{Var}(X_2) = \Sigma(x_{2i} - \mu_2)^2 / (n-1)$
$\mathrm{Cov}(X_1,X_2) = \Sigma(x_{1i}-\mu_1)(x_{2i}-\mu_2) / (n-1)$

Computed Values

$\mathrm{Var}(X_1) = 114/3 = 38$
$\mathrm{Var}(X_2) = 114/3 = 38$
$\mathrm{Cov}(X_1,X_2) = -114/3 = -38$

Covariance Matrix C

$$\begin{array}{cc} & \begin{array}{cc} X_1 & X_2 \end{array} \\ \begin{array}{c} X_1 \\ X_2 \end{array} & \begin{bmatrix} 38 & -38 \\ -38 & 38 \end{bmatrix} \end{array}$$

Negative covariance → X_1 and X_2 move in opposite directions

Step 3: Eigenvalues of Covariance Matrix

Solve $\det(C - \lambda I) = 0$ to find the eigenvalues

Characteristic Equation

$|C - \lambda I| = 0$

$$\begin{vmatrix} 38-\lambda & -38 \\ -38 & 38-\lambda \end{vmatrix} = 0$$

Solving

$(38-\lambda)^2 - (38)^2 = 0$

$\lambda^2 - 76\lambda + 1444 - 1444 = 0$

$\lambda(\lambda - 76) = 0$

Eigenvalue Results

$\lambda_1 = 76$

LARGER eigenvalue — first principal component

$\lambda_2 = 0$

Smaller eigenvalue — second principal component

$\lambda_1 = 76$ captures ALL the variance → we only need 1 principal component!

Step 4: Compute Eigenvectors

Solve $(C - \lambda_1 I)v = 0$ for the eigenvector corresponding to $\lambda_1 = 76$

System of Equations ($\lambda_1 = 76$)

$(38-76)v_1 + (-38)v_2 = 0$

$-38v_1 - 38v_2 = 0$

$-38v_1 = 38v_2$

$v_1 = -v_2$

Normalizing the Eigenvector

Length of X_1:

$\|X_1\| = \sqrt{(1^2 + (-1)^2)}$

$\|X_1\| = \sqrt{2} \approx 1.4142$

Unit eigenvector e_1:

$e_1 = X_1 / \|X_1\| = [1/\sqrt{2}, -1/\sqrt{2}]$

Taking $t = 1$: $X_1 = [1, -1]$ (un-normalized)

$e_1 = [0.707, -0.707]$

$e_2 = [0.707, 0.707]$

← PC1 (use this)

← PC2 (λ_2=0, drop)

Step 5: First Principal Components

Projection Formula

$$\mathbf{y} = \mathbf{e}_1^T \cdot \mathbf{x}_k$$
$$= 0.707 \cdot x_{1k} + (-0.707) \cdot x_{2k}$$
$$= 0.707 \cdot (x_{1k} - x_{2k})$$

Example: Sample 1 (x_1=4, x_2=11)

$$y_1 = 0.707 \times 4 + (-0.707) \times 11$$
$$= 2.828 - 7.777$$
$$= \mathbf{-4.305}$$

All Projected Values

Sample	X_1	X_2	PC_1
1	4	11	**−4.305**
2	8	4	**3.536**
3	13	5	**5.657**
4	7	14	**−5.091**

2D data successfully reduced to 1D!

Step 6 : Geometric Meaning

Project data onto e_1 axis — each point's coordinate on e_1 IS its PC_1 value

1 Original 2D Data

Four points scattered in the X_1–X_2 plane. Each needs 2 numbers to describe it.

2 Shift to Center

Move the origin to the mean point [8, 8.5]. Axes now pass through the data center.

3 Rotate to e_1

Align the new axis with eigenvector e_1. The e_1-axis captures the maximum variance.

4 Drop Perpendiculars

Project each point onto the e_1-axis. Each projection is one 1D coordinate.

1D result: { −4.305, 3.536, 5.657, −5.091 } — each a single number on the e_1-axis

Summary

1 **Calculate Mean**

$\mu_1=8,\ \mu_2=8.5$

2 **Covariance Matrix**

$C = [[38,-38],[-38,38]]$

3 **Eigenvalues**

$\lambda_1=76,\ \lambda_2=0$

4 **Eigenvectors**

$e_1=[0.707,-0.707]$

5 **Project onto PC_1**

$y = e_1^T \cdot x_k$

6 **1D Result**

$\{-4.31, 3.54, 5.66, -5.09\}$

PCA transforms correlated high-dimensional data into uncorrelated lower dimensional components.

Inclass Activity - Colab

https://colab.research.google.com/drive/1F1SvTg7IrD6sLwmZ3vpyTN443MwHdxqd?usp=sharing

Thank You

Reference:

MIT- 940 Introduction to Neural Computation
https://vtupulse.com/machine-learning/principal-component-analysis-in-machine-learning/